William Randolph Oak

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AI-Era Professional Summary

Senior Software Engineer and Systems Architect with 30+ years of experience delivering distributed, real-time, and fault-tolerant systems across multiple technology revolutions. Having lived through the Computer Revolution, the rise of the Internet, and the Dot-Com boom, I recognize hype cycles and focus on engineering fundamentals that produce lasting value.

Formally trained in Machine Learning through Stanford University (Andrew Ng), I use AI as a force multiplier—accelerating research, mastering new technologies quickly, and validating complex architectures.

Recent work includes architecting a Kubernetes-native, self-healing, horizontally scalable SIP-based communication platform with distributed storage, TypeScript microservices, a React UI, and a secure single-source-of-truth API.

Career Highlights Overview

- Architected a full distributed communication platform with Kubernetes, distributed storage, SIP routing, and a modern TypeScript/React stack.
- Rebuilt a proprietary legacy telecom system from the ground up, delivering a scalable, offline-capable, future-ready architecture.
- Established complete engineering foundations—workflow, documentation, CI/CD requirements, testing strategy, and team structure—from a zero-infrastructure start.
- Designed a resilient storage model using replicated configuration volumes and distributed data volumes with RAID-like durability.
- Leveraged AI to rapidly master new technologies, accelerate architectural validation, and extend engineering throughput.
- Modernized the entire platform, migrating backend services and the configuration UI from PHP to a unified TypeScript ecosystem.

Professional Experience

Senior Software Engineer / Systems Architect — Harding (2024–2025)

Served as principal architect for the complete replacement of the legacy intercom system. Designed a Kubernetes-native, distributed, self-healing, offline-capable communication platform integrating SIP routing, distributed storage, TypeScript microservices, and a React-based UI. Established all development practices, documentation standards, and CI/CD requirements from a zero-infrastructure starting point.

Key Contributions

- Architected the full platform: cluster topology, distributed storage, SIP engine integration, microservices, APIs, and UI.
- Built a working Kubernetes-based cluster with GlusterFS and encrypted, self-healing node behavior.
- Designed the distributed SQL model and cluster interactions using CockroachDB.
- Developed TypeScript listeners for FreeSWITCH ESL, Hostport, and legacy protocols.
- Created a secure REST API with WebSocket-based real-time propagation.
- Migrated backend services and UI from PHP to a unified TypeScript/React stack.
- Reverse-engineered the legacy DXL system to ensure backward compatibility.
- Defined CI/CD architecture and engineering standards.
- Used AI to accelerate research, troubleshooting, and architectural decision-making.

Senior Software Engineer — Veritree (2023)

Implemented STRAPI configuration architecture, enabling robust client-facing content control. Practiced TDD with PHPUnit and contributed to GitHub→AWS deployment workflows.

Senior Software Engineer — Zaui (2019–2023)

Led feature development for a major SaaS platform, mentored developers, introduced Agile practices, and contributed to CI/CD operations. TDD using PHPUnit.

Senior Software Engineer — XenCALL (2016–2019)

Developed real-time telecom/CRM features integrating PHP, JavaScript, MySQL, Memcached, WebSockets, and Asterisk. Led Agile team and contributed to AWS-based deployments.

Programmer & Senior Network Consultant — Northshore Driving School (2000–2016)

Developed scheduling systems, maintained servers, migrated infrastructure to cloud environments, and built backend CMS features.

Programmer & Senior Network Specialist — Website-in-a-Box (2007–2016)

Built PERL-based website builder, maintained servers, backups, and reliability systems.

Team Lead — Mission:IMPossible (2011–2012)
Architected backend CMS for UFCW Local 1518, including QR code automation, authentication modules, and content management tools.

Core Competencies

Foundational Engineering Competencies

- End-to-end systems architecture for distributed, fault-tolerant, and real-time platforms
- High-availability data modeling, distributed consistency strategies, and schema governance
- Cluster orchestration, resilience engineering, capacity planning, and performance scaling
- Telecom and event-driven system design, including SIP-based signaling and protocol translation
- Secure system interface design: REST architectures, real-time event propagation, and cross-service contracts
- Full-stack solution design spanning backend microservices and modern, maintainable UI frameworks
- Linux-based systems engineering, network topology design, storage reliability, and infrastructure hardening
- Technical leadership: establishing engineering standards, workflows, and multi-disciplinary team structure
- Architecture for highly constrained environments, including offline, air-gapped, and multi-site deployments

AI-Era Amplified Competencies

- Rapid acquisition of complex technical domains through AI-accelerated learning workflows
- AI-augmented debugging, failure analysis, and architecture validation for distributed systems
- AI-supported design documentation, system modeling, and technical communication
- Machine-learning–based evaluation of engineering tradeoffs and feasibility constraints
- LLM-enabled expansion of cross-domain engineering capacity and delivery velocity

Technology Mastery Overview

Distributed Systems & Orchestration:
Kubernetes (secure, multi-node orchestration), GlusterFS (replicated/distributed volumes), CockroachDB (distributed SQL), resilience, failover, and offline-capable architecture.

Telecom & Real-Time Systems:
FreeSWITCH ESL, Asterisk, SIP signaling, real-time event pipelines, legacy protocol translation (mixed-protocol environments).

Backend Architecture:
Node.js/TypeScript microservices, REST and WebSocket system design, single-source-of-truth API frameworks, 20+ years LAMP expertise.

Frontend Architecture:
React (TypeScript), configuration/admin UIs, scalable UI patterns, and UX collaboration.

CI/CD & Engineering Infrastructure:
Pipeline requirements design, secure deployment models, Git workflow strategy, OpenAPI/Swagger documentation standards.

Systems & Networking:
Linux systems engineering (20+ years), DNS/DHCP/BIND, cluster networking, encrypted service meshes, performance and reliability hardening.

AI & Machine Learning:
Stanford ML foundations, AI-assisted research, architectural validation, rapid domain acquisition through LLM-driven workflows.

Programming Languages:
TypeScript, JavaScript, PHP, Bash, PERL, Python, SQL, C/Assembly (historical).

Education & Certifications

Machine Learning — Stanford University (via Coursera)
Instructor: Andrew Ng, Associate Professor, Stanford University
Verified Certificate • March 12, 2019 • Grade: **97.25%** • ~59 hours
Identity-verified credential:
https://www.coursera.org/account/accomplishments/verify/DEMXPANGXTAH
Instructor profile: https://hai.stanford.edu/people/andrew-ng

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References

Tim Pippus — CEO, Mission:Impossible
Dade Brandon — CTO, XenCALL
Brett Whitehead — CTO, Zaui
Full contact information available upon request.